ORIGINAL ARTICLES

ACE Inhibitor Mediated Reductions in Renal Size and Microalbuminuria in Normotensive, Diabetic Subjects

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ABSTRACT

Recent studies utilizing converting enzyme inhibitors (CEI) in diabetic rats document reductions in both renal hypertrophy and albuminuria. Four separate clinical studies in normotensive patients with diabetes demonstrate reduction of microalbuminuria with CEIs independent of blood pressure reduction. The present pilot study examines the results of reducing an elevated glomerular filtration rate on changes in renal size and microalbuminuria in normotensive, hyperfiltering insulin-dependent diabetic (IDDM) patients. Fifteen IDDM patients were randomized to either placebo or the CEI, lisinopril. Dosage of lisinopril was titrated over 3 months to reduce glomerular filtration rate (GFR) to ≤ 2.33 mL/sec. Evaluation at 18 months demonstrated the lisinopril group had a marked reduction in renal size (16.9 ± 1.1, baseline versus 12.8 ± 0.9 cm, 18 months; $p < 0.05$) and microalbuminuria (92 ± 11 μg/min, baseline versus 23 ± 26 μg/min, 18 months; $p < 0.05$). No change in renal size was noted in the placebo group (15.4 ± 0.8, baseline versus 14.9 ± 0.7 cm, 18 months; NS) and albuminuria increased (118 ± 15 μg/min, baseline versus 293 ± 32 μg/min, 18 months; $p < 0.05$). Mean arterial pressure at 18 months was significantly lower in the lisinopril group compared to placebo (102 ± 4, placebo versus 87 ± 6 mm Hg, CEI, $p < 0.05$). This study supports previous animal studies that document reductions in both microalbuminuria and renal size by a CEI. The overall impact of these findings on preservation of renal function cannot be assessed, however, in this short-term study. (Journal of Diabetes and Its Complications 8;1:2-6, 1994.)

INTRODUCTION

Clinical studies in both insulin-dependent (IDDM) and non-insulin-dependent (NIDDM) diabetic patients document that persistent arterial pressure reduction to levels below 140/90 mm Hg slows the rate of decline in glomerular filtration rate.[1-4] Recent studies in both normotensive and hypertensive diabetic patients also report differential effects on declines in renal function by certain classes of antihypertensive agents.[1,4-10] These observations are not surprising given the data garnered from animal studies. Numerous studies in animal models of diabetes demonstrate that control of arterial pressure preserves renal function and reduces glomerulosclerosis.[1-4] Moreover, these studies also document that certain classes of antihypertensive agents have specific morphologic effects unrelated to

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TABLE 1. BASELINE CHARACTERISTICS OF TYPE I DIABETIC POPULATION[a] STUDIED

	Placebo	Lisinopril	Control[b]
n	7	8	15
Age	25 ± 6	28 ± 7	29 ± 6
Sex (M/F)	4/3	3/5	10/5
Duration of diabetes (yr)	7 ± 3	9 ± 2	8 ± 4
Body weight (kg)	73 ± 8	76 ± 5	81 ± 7
SBP (mm Hg)	126 ± 5	131 ± 6	137 ± 8*
DBP (mm Hg)	78 ± 3	79 ± 4	88 ± 7*
Cr_{cl} (mL/sec)	3.07 ± 0.04	3.11 ± 0.08	2.88 ± 0.09
Smoking history (%)	19	25	20
Renal size[c] (cm)	15.3 ± 0.8	16.8 ± 0.9	14.9 ± 1.2

SBP, systolic blood pressure; DPB, diastolic blood pressure; Cr_{cl}, creatinine clearance.

[a] *All had positive family history for cardiovascular disease or hypertension.*

[b] *Eight of 15 were on combinations of thiazide diuretics with the β blocker atenolol (50-100 mg); four of 15 were on β blocker alone (atenolol 50-100 mg); and three of 15 were on thiazide diuretic alone (12.5 - 25 mg hydrochlorothiazide). These data reflect information from patient records retrieved when patients were selected as controls.*

[c] *Determined by ultrasound within 3 years of initial diagnosis of diabetes; *$p < 0.05$ significantly different than placebo or lisinopril.*

blood pressure reduction.[13,14] Specifically, converting enzyme inhibitors are known to attenuate increases in albuminuria and both the renal hemodynamic (glomerular capillary pressure) and morphologic changes (glomerular hypertrophy, mesangial matrix expansion) associated with diabetes.[4,12-14]

Animal studies also demonstrate that antihypertensive agents that reduce both glomerular pressure as well as volume result in better morphologic preservation over agents that only reduce volume.[13,14] Moreover, these studies report that converting enzyme inhibitors generally reduce renal size.[11-15] This is not always true, however, because in some animal experiments CEIs did not markedly affect renal size.[11,15]

Similar to animal models of diabetes, patients exhibit elevations in glomerular filtration rate in the early stages of either insulin-dependent or non-insulin-dependent diabetes.[4,16-18] Although some studies have used converting enzyme inhibitors to evaluate changes in albuminuria among normotensive diabetic subjects,[5-9] no study in human subjects has evaluated the effects of these agents on changes in renal size or microalbuminuria in association with a reduction in the pre-existing elevated glomerular filtration rate, i.e., hyperfiltration. The present study was undertaken to assess whether or not a reduction of an elevated glomerular filtration rate in IDDM patients is associated with a reduction in both renal size and microalbuminuria.

METHODS

Twenty-two normotensive IDDM patients with elevated glomerular filtration rates, i.e., greater than 2.5 mL/sec and a positive family history for cardiovascular disease and/or hypertension were selected for the study. Fifteen patients successfully completed the study and form the basis of this report. The other seven were terminated for one of the following reasons: four of the seven were noncompliant with either blood sugar control, diet, or taking study drug; one subject dropped out at 5 months for symptoms of dizziness and orthostasis; and the remaining two individuals elected to stop the study at 6 and 8 months due to schedule conflicts with their jobs.

The baseline demographic and arterial pressure characteristics of the population are summarized in **Table 1**. All subjects were placed on a 120 mEq sodium, 1.0 g per kg protein diet and periodically monitored throughout the duration of the study. An illustrated protocol of the study is shown in **Figure 1**. Patients were prospectively randomized to either the converting enzyme inhibitor, lisinopril (n = 8), or a placebo (n = 7). A drug-titration period ensued over a 3-month period to reduce creatinine clearance to less than or equal to 2.33 mL per second. During this titration, a standing arterial pressure of 110/60 mm Hg or less, or presence of symptoms excluded the patient from the study. Once the goal glomerular filtration rate was achieved, data collection ensued over the next 18 months. Data from these patients were compared to a control group (n = 15) of IDDM patients who were age and sex matched with a similar duration of IDDM. The control group, however, had mild (Stage I) hypertension and were receiving either a diuretic, β blocker, or both for approximately the same duration as the other groups studied. This group was retrospectively selected and not randomized to different blood pressure medications. Both study groups had formal assessment of the following parameters: fasting blood sugars, hemoglobin A_{1C}, glomerular filtration rate by diethylenetriaminepentacetic acid, [^{99}Tc]DTPA, renal blood flow by paraaminohippurate, [^{131}]PAH, 24-h

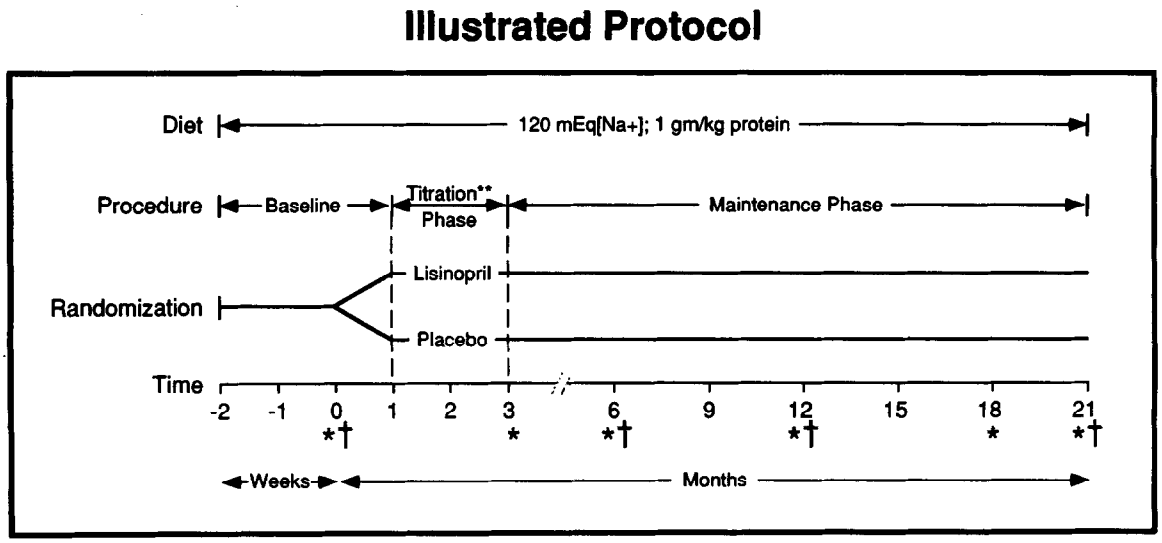

FIGURE 1 *An illustrated protocol of the study. *Measurements of renal hemodynamics, renal ultrasound. **Dosages of either placebo or lisinopril were increased by 20 mg increments every 2 weeks.*

urine for microalbumin, sodium, and urea nitrogen, as well as renal ultrasound for renal size. The control group also had the same evaluations of renal function performed at the same time intervals as the study patients.

Measurement of glomerular filtration rate and renal blood flow were assessed using a constant infusion of isotopes and measuring plasma/urine disappearance curves. Blood pressures were measured weekly in triplicate by the same observer using a Dinemapp arterial pressure measuring device. All other parameters, with the exception of hemoglobin A_{1C} and renal ultrasound, were measured bimonthly. Hemoglobin A_{1C} and renal ultrasound were performed every 3 months. Renal ultrasound examinations were performed on all subjects, including control subjects, by the same technician and read by the same radiologist. The radiologist was blinded as to randomization of the patients. Renal ultrasound measured the maximum distance between points in two dimensional planes.

RESULTS

No significant differences between study groups were noted over the 18-month period of study with regard to hemoglobin A_{1C} (8.5 ± 0.5%, placebo versus 8.2 ± 0.4%, lisinopril; $p = 0.48$) or fasting blood sugar control (9.5 ± 0.4, placebo versus 9.3 ± 0.5 mM/L lisinopril; $p = 0.51$). In addition, dietary compliance, as measured by 24-h urine sodium and urea nitrogen values, and arterial pressure control as measured by an average of three readings with a Dinemapp device were similar, **Table 2**. **Figure 2** demonstrates the overall systemic, vascular, and renal hemodynamic profiles as well as changes in renal size over the 18-month study. Renal size significantly decreased only in the lisinopril group compared to either placebo (24 ± 3% lisinopril versus 3.4 ± 2.2% placebo; $p < 0.05$), or control groups (24 ± 3% lisinopril versus 5.3 ± 2.8% control; $p < 0.05$) at 18 months. Significant reductions in renal size were noted as early as 12 months in the lisinopril group (13.7 ± 2.3% lisinopril versus 1.1 ± 0.9%, placebo; $p < 0.05$). The mean dose of lisinopril required to reduce glomerular filtration rate was 78 ± 8 mg/day. The relationship between increasing dosages of lisinopril and changes in glomerular filtration rate are summarized in **Table 3**. Glomerular filtration rate was not reduced until lisinopril dose was at least 60 mg per day. Microalbuminuria normalized in all but two patients at 12 months, **Figure 2**. This normalization in microalbuminuria persisted to the end of the study, **Figure 2**.

TABLE 2. DIETARY COMPLIANCE AND ARTERIAL PRESSURE MEASUREMENTS AT 18 MONTHS

	Placebo	Lisinopril	Control[a]
n	7	8	15
Body weight (kg)	71 ± 5	73 ± 4	78 ± 6
SBP (mm Hg)	134 ± 4*	115 ± 4	122 ± 5
DBP (mm Hg)	86 ± 3*	72 ± 2	75 ± 3
24-h $U_{Na}V$ (mmol)	112 ± 8	123 ± 10	106 ± 12
$U_{urea}V$ (mmol/d)	11 ± 3	10 ± 2	13 ± 4

SBP, systolic blood pressure; DBP, diastolic blood pressure; $U_{Na}V$, urine sodium volume; $U_{urea}V$, urine urea nitrogen value.

[a] *Eight of 15 were on combinations of thiazide diuretics with the β blocker atenolol (50–100 mg); four of 15 were on β blocker alone (atenolol 50–100 mg); and three of 15 were on thiazide diuretic alone (12.5–25 mg hydrochlorothiazide). These data reflect information from patient records retrieved when patients were selected as controls.*

* *$p < 0.05$ compared to lisinopril or control.*

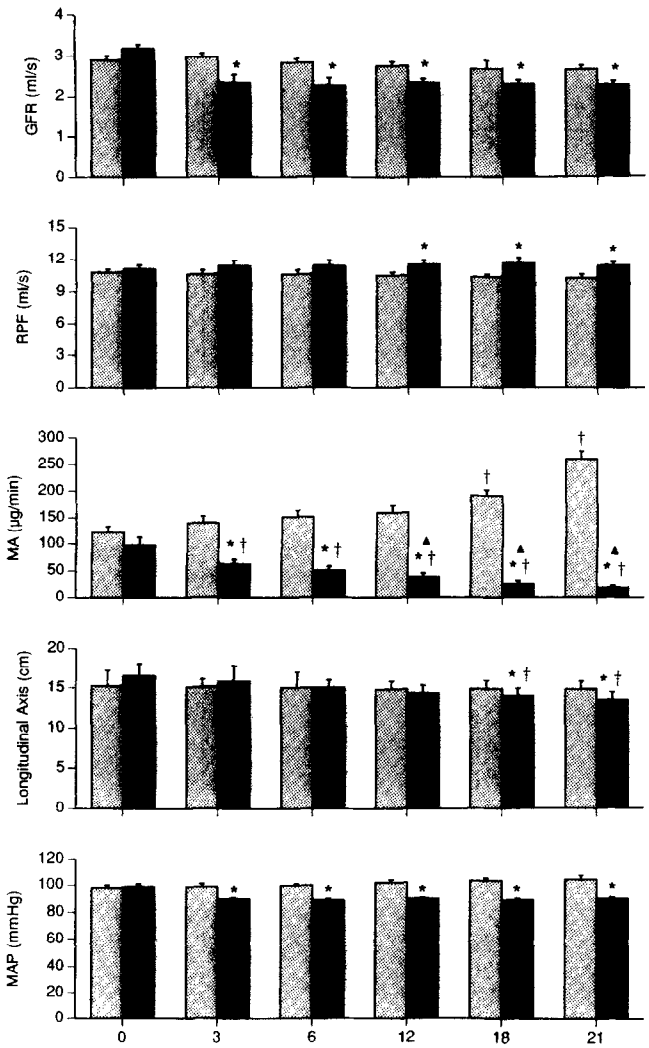

FIGURE 2 *The effects of lisinopril on renal size, microalbuminuria, and renal hemodynamics over an 18-month period. Abbreviations: GFR, glomerular filtration rate; RPF, renal plasma flow; MA, microalbuminuria; Longitudinal Axis, renal size by ultrasound; and MAP, mean arterial pressure. Note: 0 is baseline prior to randomization; 3 is the first reading following titration to lower GFR. Shaded bars, placebo; dark bars, lisinopril. *p < 0.05 compared to baseline (Time 0) reading.*

DISCUSSION

Use of a converting enzyme inhibitor in normotensive IDDM patients is known to reduce microalbuminuria.[5-9,17] No study, however, has assessed the possible association between reduction of elevated glomerular filtration rate and decrease in renal size. We demonstrate a reduction in both renal size and microalbuminuria following titration of a converting enzyme inhibitor to doses that reduce above normal glomerular filtration rates in normotensive diabetic subjects. Specifically, we demonstrate that the converting enzyme inhibitor, lisinopril, substantially reduces microalbuminuria and renal size as early as 1 year after the initiation of treatment.

It must be noted that a reduction in glomerular filtration rate and concomitant decrease in renal size are associated but not necessarily causally related. Several other factors could contribute to this reduction in renal size. These include reduction in arterial pressure or an attenuated production or effects of various vasoconstrictor growth factors that include angiotensin II, norepinephrine, endothelin, and others. Moreover, to our knowledge, only one study has assessed the response of various growth factors in the internal milieu of an elevated glomerular capillary pressure.[20] Our study, unfortunately, could not address these issues in a specific way.

Reduction in arterial pressure would seem to be a plausible answer to explain the reduction in both renal size and microalbuminuria. We found, however, no reduction in renal size or albuminuria, in spite of similar blood pressure reduction, in our control group. Moreover, recent animal studies support the notion that similar reductions in arterial pressure with different classes of antihypertensive medications do not correlate with similar reductions in glomerular capillary pressure, renal size or progression of the morphologic changes seen with diabetes.[11-14] Taken together, these data support the concept that converting enzyme inhibitors have renal effects independent of arterial pressure.[8]

Most antihypertensive medications can initially reduce microalbuminuria simply by effecting large reductions in arterial pressure.[8,20] These reductions, however, are not sustained with the majority of medications.[4] Converting enzyme inhibitors, however, have been shown to have effects on the glomerular filtration barrier that directly affects protein leaks from the glomerulus. The effects of converting enzyme inhibitors also appear to be independent of arterial pressure reduction.[4,21,22] Our study, unfortunately, did not measure these parameters and utilized high doses of

TABLE 3. TITRATION OF THE CONVERTING ENZYME INHIBITOR, LISINOPRIL, TO REDUCE ARTERIAL PRESSURE AND CREATININE CLEARANCE

Dosage (mg/d)	GFR (mL/sec)	MAP (mm Hg)
0	3.1 ± 0.06	99 ± 2
20	3.04 ± 0.05	98 ± 3
40	2.92 ± 0.09	96 ± 3
60	2.57 ± 0.08*	94 ± 2*
80	2.34 ± 0.09**	94 ± 2*
100	2.29 ± 0.11**	91 ± 4**

GFR, glomerular filtration rate; MAP, mean arterial pressure.

** p < 0.05 compared to 20 mg/d; ** p < 0.05 compared to 60 mg/d dosage.*

lisinopril to reduce glomerular filtration rate. Thus, we can only confirm previous studies that report reductions in albuminuria.

We conclude that reduction of a preexisting elevation in glomerular filtration rate can lead to a reduction in renal size and microalbuminuria. While we cannot exclude a blood pressure effect, we feel that further studies need to be performed to address the issue of reduction in renal size on the natural history of diabetic nephropathy.

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REFERENCES

1. Bakris GL, Barnhill BW, Sadler R. Treatment of arterial hypertension in diabetic humans: Importance of therapeutic selection. *Kidney Int* 41:912–929, 1992.
2. Parving HH. Impact of blood pressure and antihypertensive treatment on incipient and overt nephropathy, retinopathy and endothelial permeability in diabetes mellitus. *Diabetes Care* 14:260–269, 1991.
3. Parving HH. The impact of hypertension and anti-hypertensive treatment on the course and prognosis of diabetic nephropathy. *J Hypertens* 8(suppl 7):S187–S191, 1990.
4. Bakris GL. Calcium abnormalities and the diabetic, hypertensive patient: Implications for renal preservations, *in* Epstein M (ed). *Calcium Antagonists in Clinical Medicine*. Philadelphia, Hauley and Belfus, 1992, pp. 367–389.
5. Marre M, Chatellier G, Leblanc H, Guyene TT, Menard T, Passa P. Prevention of diabetic nephropathy with enalapril in normotensive diabetics with microalbuminuria. *BMJ* 297:1092–1095, 1988.
6. Parving HH, Hommel E, Damkjaer Nelsen M, Giese J. Effect of captopril on blood pressure and kidney function in normotensive insulin dependent diabetics with nephropathy. *BMJ* 299:533–536, 1989.
7. Stornello M, Valvo EV, Scapellato L. Angiotensin converting enzyme inhibition in normotensive type II diabetics with persistent mild proteinuria. *J Hypertens Suppl* 7:S464–S466, 1988.
8. Bakris GL. Angiotensin converting enzyme inhibitors and progression of diabetic nephropathy. *Ann Intern Med* 118:643–644, 1993.
9. Marre J, Chatellier G, Leblanc H, Guyene TT, Menard J, Passa P. Prevention of diabetic nephropathy in normotensive insulin dependent diabetic patients with microalbuminuria. *BMJ* 297:1092–1097, 1988.
10. Slataper R, Vicknair N, Sadler R, Bakris GL. Comparative effects of different antihypertensive treatments on progression of diabetic renal disease. *Arch Intern Med* 153:973–980, 1993.
11. Anderson S, Rennke HG, Garcia DL, Brenner BM. Short and long term effects of antihypertensive therapy in diabetic rats. *Kidney Int* 36:526–536, 1989.
12. Brown S, Walton C, Crawford P, Bakris GL. Renal response to angiotensin converting enzyme inhibition or calcium channel blockage in diabetic beagles. *Kidney Int* 43:1210–1218, 1993.
13. Gaber L, Walton C, Brown S, Bakris GL. Effects of converting enzyme inhibition and calcium antagonism alone and in combination on glomerular morphology in diabetic dogs [Abstract]. *J Am Soc Nephrol* 3:428, 1992.
14. Anderson S, Rennke HG, Brenner BM. Nifedipine versus fosinopril in uninephrectomized diabetic rats. *Kidney Int* 41:891–897, 1992.
15. Hajinazarian M, Cosio FG, Nahman NS. Captopril attenuates the organomegoly and glomerular hypertrophy of diabetes [Abstract]. *J Am Soc Nephrol* 3:758, 1992.
16. Vora JP, Dolben J, Dean JD, Thomas D, Williams JD, Owens OR, Peters JR. Renal hemodynamics in newly presenting non-insulin dependent diabetes mellitus. *Kidney Int* 41:829–835, 1991.
17. Mogensen C, on behalf of the European Microalbuminuria Captopril Study Group. Captopril delays progression to overt renal disease in IDDM [Abstract]. *J Am Soc Nephrol* 3:643, 1992.
18. Mogensen CE. Natural history of renal functional abnormalities in human diabetes mellitus: From normoalbuminuria to incipient and overt nephropathy. *Contemp Issues Nephrol* 20:19–49, 1989.
19. Rosenberg ME, Hostetter TH, with technical assistance of Kren S and Chmielewski D. Effect of angiotensin II and norepinephrine on early growth response genes in the rat kidney. *Kidney Int* 43:601–609, 1993.
20. Melbourne Diabetic Nephropathy Study Group. Comparison between perindopril and nifedipine in hypertensive and normotensive diabetic patients with microalbuminuria. *BMJ* 302:210–216, 1991.
21. Scandling JD, Myers BD. Glomerular size selectivity and microalbuminuria in early diabetic glomerular disease. *Kidney Int* 41:840–846, 1992.
22. Loon N, Shemesh O, Morelli E, Myers BD. Effect of angiotensin II infusion on the human glomerular filtration barrier. *Am J Physiol* 257:F608–614, 1989.